Exit Exam Practice Questions

60 Additional Multiple Choice Questions
Organized by Theme

Software Engineering and Computing Technology
National Exit Examination Preparation

Instructions: Select the best answer for each question.
Answers and detailed explanations provided at the end.

Theme 1: Problem Analysis and Programming

Fundamentals of Programming

Q1: What is the first step in the program development process?

A) Writing code
B) Problem analysis and understanding requirements
C) Testing the program
D) Debugging errors

Q2: Which of the following is NOT a basic programming construct?

A) Variables
B) Control structures (if-else, loops)
C) Functions/Methods
D) Hardware components

Q3: What does the 'try-catch' block do in Java?

A) Repeats code until a condition is true
B) Handles exceptions (errors) gracefully
C) Tests whether a statement is true
D) Declares a new variable

Q4: In Java, what is the maximum value for an 'int' variable?

A) 100
B) 1,000
C) 2,147,483,647
D) Unlimited

Q5: What is the purpose of a 'return' statement in a method?

A) To end the program
B) To send a value back to the caller and exit the method
C) To skip to the next line
D) To reset variables

Q6: How do you read a single character from user input in Java?

A) Scanner.readInt()
B) Scanner.next().charAt(0)
C) Scanner.readChar()
D) Scanner.getChar()

Q7: What is the difference between pass-by-value and pass-by-reference?

A) Pass-by-value is faster
B) In pass-by-value, changes don't affect original; in pass-by-reference, they do
C) Pass-by-reference is only for objects
D) They're the same thing

Q8: What is the purpose of testing in the program development process?

A) To ensure the program is fast
B) To find and fix errors before deployment
C) To make the code look neat
D) To improve variable names

Q9: What is a modular program?

A) A program that runs on multiple computers
B) A program divided into separate, independent modules/functions
C) A program with many lines of code
D) A program that uses multiple classes

Q10: What is the difference between a while loop and a do-while loop?

A) While loops are faster
B) Do-while executes at least once; while checks condition first
C) Do-while is only for object iteration
D) There's no difference

Q11: What is the purpose of a constructor in a Java class?

A) To delete objects
B) To initialize object properties when created
C) To define methods
D) To declare variables

Q12: How many times does this loop execute? for(int i=0; i<5; i++) { }

A) 0 times
B) 4 times
C) 5 times
D) 6 times

Q13: What is the time complexity of inserting an element at the beginning of a linked list?

A) O(1)
B) O(n)
C) O(log n)
D) $O(n^2)$

Q14: Which data structure is best for implementing a browser's back button functionality?

A) Queue
B) Stack
C) Tree
D) Hash Table

Q15: In a heap, what relationship must exist between parent and child nodes?

A) No specific relationship
B) Parent < child (min heap) or parent > child (max heap)
C) Parent always = child
D) Child must be twice the parent

Q16: What is the space complexity of merge sort?

A) O(1)
B) O(log n)
C) O(n)
D) $O(n^2)$

Q17: Which of the following best describes a graph's adjacency list representation?

A) A 2D array showing connections between vertices
B) A list for each vertex containing its adjacent vertices
C) A linked list of all graph edges
D) A single array of all vertices

Q18: What is the worst-case time complexity of quicksort?

A) O(n)
B) O(n log n)
C) $O(n^2)$
D) O(2^n)

Q19: In a doubly-linked list, what additional pointer does each node have compared to singly-linked?

A) Parent pointer
B) Previous pointer
C) Random pointer
D) Child pointer

Q20: What is the primary advantage of a hash table over a binary search tree?

A) Hash tables maintain sorted order
B) Hash tables have O(1) average search time vs O(log n)
C) Hash tables use less memory
D) Hash tables support range queries

Q21: What is the time complexity of inserting an element at an arbitrary position in an unsorted array?

A) O(1)
B) O(log n)
C) O(n)
D) $O(n^2)$

Q22: In a complete binary tree with n nodes, what is the height?

A) n
B) log(n)
C) n/2
D) sqrt(n)

Q23: What is the main drawback of bubble sort?

A) Difficult to understand
B) Requires extra memory
C) Very slow even for small arrays - $O(n^2)$
D) Cannot sort objects

Q24: Which algorithm is generally best for small arrays of random data?

A) Merge Sort
B) Quick Sort
C) Insertion Sort
D) Bubble Sort

Q25: Which principle emphasizes hiding internal details of an object?

A) Inheritance
B) Polymorphism
C) Encapsulation
D) Abstraction

Q26: What keyword is used in Java to inherit from a class?

A) inherit
B) extends
C) implements
D) inherits

Q27: Can an abstract class be instantiated?

A) Yes, always
B) No, never
C) Only if it has abstract methods
D) Only from subclasses

Q28: What is method overloading?

A) Calling a method multiple times
B) Multiple methods with same name but different parameters
C) A method calling itself
D) Using methods from parent class

Q29: What is method overriding?

A) Defining a method with the same name in a subclass
B) Calling a method multiple times
C) Changing method visibility
D) Creating a new method

Q30: What is the 'super' keyword used for?

A) To declare a variable
B) To call parent class methods or constructors
C) To end a program
D) To create a new object

Q31: An interface can have concrete method implementations (Java 8+). True or False?

A) False, interfaces are pure abstraction
B) True, using 'default' and 'static' keywords
C) True, but only in main class
D) False, only abstract classes can

Q32: Which SOLID principle advocates depending on abstractions, not concrete classes?

A) Single Responsibility
B) Open-Closed
C) Dependency Inversion
D) Interface Segregation

Q33: What is the purpose of the 'this' keyword?

A) To refer to the parent class
B) To refer to the current object instance
C) To create a new object
D) To end a method

Q34: A class can implement multiple interfaces. True or False?

A) False
B) True
C) Only if they don't conflict
D) Only in abstract classes

Q35: What is the correct order of OOP principles from least to most restrictive access?

A) public, protected, private, default
B) default, public, protected, private
C) private, protected, default, public
D) public, default, protected, private

Q36: What is a static method?

A) A method that doesn't change
B) A method belonging to the class, not instances (called via class name)
C) A method that's always private
D) A method without parameters

Q37: What does REST stand for?

A) Rapid Execution Service Technology
B) Representational State Transfer
C) Resource Exchange System Token
D) Remote Execution Service Tool

Q38: Which HTTP method is idempotent and should be used to retrieve data?

A) POST
B) PUT
C) GET
D) DELETE

Q39: What is the purpose of CSS in web development?

A) Define website structure
B) Add interactivity with events
C) Style and layout website appearance
D) Manage server operations

Q40: In Android, which lifecycle method is called first?

A) onStart()
B) onCreate()
C) onResume()
D) onPause()

Q41: What is the difference between an Intent and an Intent Filter?

A) They're the same
B) Intent is a message; Intent Filter declares what intents an app can handle
C) Intent Filter is used in Java; Intent in XML
D) Intent Filters are deprecated

Q42: How do you pass data between activities in Android?

A) Using global variables
B) Using Bundle and Intent.putExtra()
C) Using static fields
D) Using shared preferences directly

Q43: What is a Fragment in Android?

A) Part of an activity that can be reused
B) A complete activity alternative
C) A service component
D) A type of Intent

Q44: What is JSON primarily used for in web services?

A) Database structure
B) Server operating system
C) Data exchange format between client and server
D) Client-side styling

Q45: What does DOM (Document Object Model) represent?

A) The database structure
B) The tree structure of HTML elements in a page
C) The server configuration
D) The CSS structure

Q46: What is responsive web design?

A) Websites that respond to user input
B) Designing websites that work well on all screen sizes
C) Fast-loading websites
D) Websites that connect to databases

Q47: What is the purpose of a web framework (like Spring, Django, Laravel)?

A) To draw graphics on websites
B) To provide reusable components for building web applications
C) To manage databases only
D) To encrypt data

Q48: What is cross-site scripting (XSS)?

A) Multiple scripts on one page
B) A security vulnerability where malicious scripts are injected into web pages
C) Using multiple programming languages
D) Communicating between websites

ANSWER KEY

Q1: **B** | Q2: **D** | Q3: **B** | Q4: **C** | Q5: **B** | Q6: **B** | Q7: **B** | Q8: **B** | Q9: **B** | Q10: **B** |

Q11: **B** | Q12: **C** | Q13: **A** | Q14: **B** | Q15: **B** | Q16: **C** | Q17: **B** | Q18: **C** | Q19: **B** | Q20: **B** |

Q21: **C** | Q22: **B** | Q23: **C** | Q24: **C** | Q25: **C** | Q26: **B** | Q27: **B** | Q28: **B** | Q29: **A** | Q30: **B** |

Q31: **B** | Q32: **C** | Q33: **B** | Q34: **B** | Q35: **D** | Q36: **B** | Q37: **B** | Q38: **C** | Q39: **C** | Q40: **B** |

Q41: **B** | Q42: **B** | Q43: **A** | Q44: **C** | Q45: **B** | Q46: **B** | Q47: **B** | Q48: **B** |

DETAILED EXPLANATIONS

Q1: What is the first step in the program development process?

Answer: B

Explanation: Problem analysis is the first critical step where you understand what needs to be solved before writing any code.

Q2: Which of the following is NOT a basic programming construct?

Answer: D

Explanation: Hardware is not a programming construct. Variables, control structures, and functions are fundamental building blocks of programs.

Q3: What does the 'try-catch' block do in Java?

Answer: B

Explanation: Try-catch is an exception handling mechanism. The try block contains code that might throw an exception, and catch handles it.

Q4: In Java, what is the maximum value for an 'int' variable?

Answer: C

Explanation: Java's int is 32-bit signed, with max value $2^{31}-1$ = 2,147,483,647. Use 'long' for larger values.

Q5: What is the purpose of a 'return' statement in a method?

Answer: B

Explanation: Return statements exit a method and optionally send back a value to the caller based on the method's return type.

Q6: How do you read a single character from user input in Java?

Answer: B

Explanation: Use next().charAt(0) to read a single character from a Scanner object. There's no direct readChar() method.

Q7: What is the difference between pass-by-value and pass-by-reference?

Answer: B

Explanation: Pass-by-value copies the value, changes don't affect original. Pass-by-reference passes the address, changes do affect original.

Q8: What is the purpose of testing in the program development process?

Answer: B

Explanation: Testing verifies that the program works correctly, finding and fixing bugs before users experience them.

Q9: What is a modular program?

Answer: B

Explanation: Modular programming breaks a program into smaller, reusable, manageable modules/functions, improving maintainability.

Q10: What is the difference between a while loop and a do-while loop?

Answer: B

Explanation: Do-while checks the condition after executing, ensuring at least one iteration. While checks first, may never execute.

Q11: What is the purpose of a constructor in a Java class?

Answer: B

Explanation: Constructors initialize objects with initial values. They're called when 'new' creates an instance.

Q12: How many times does this loop execute? for(int i=0; i<5; i++) { }

Answer: C

Explanation: Loop runs for i=0,1,2,3,4. When i becomes 5, condition i<5 is false, so it stops. Total: 5 iterations.

Q13: What is the time complexity of inserting an element at the beginning of a linked list?

Answer: A

Explanation: Insertion at the beginning requires only updating the head pointer, regardless of list size. This is O(1) constant time.

Q14: Which data structure is best for implementing a browser's back button functionality?

Answer: B

Explanation: A stack (LIFO) is perfect for back button: push pages as you visit, pop to go back. Last visited is first returned.

Q15: In a heap, what relationship must exist between parent and child nodes?

Answer: B

Explanation: Heap property: min heap has parent ≤ children, max heap has parent ≥ children. This creates a partially ordered structure.

Q16: What is the space complexity of merge sort?

Answer: C

Explanation: Merge sort requires O(n) extra space to merge subarrays. This is the trade-off for guaranteed O(n log n) time.

Q17: Which of the following best describes a graph's adjacency list representation?

Answer: B

Explanation: Adjacency list: each vertex has a list of vertices it connects to. Space-efficient for sparse graphs.

Q18: What is the worst-case time complexity of quicksort?

Answer: C

Explanation: Worst case occurs when pivot is always the smallest/largest, creating unbalanced partitions: $O(n^2)$. Average case is O(n log n).

Q19: In a doubly-linked list, what additional pointer does each node have compared to singly-linked?

Answer: B

Explanation: Doubly-linked lists have next and previous pointers, allowing traversal in both directions.

Q20: What is the primary advantage of a hash table over a binary search tree?

Answer: B

Explanation: Hash tables achieve O(1) average case for search/insert/delete, faster than BST's O(log n). Trade-off: no ordering.

Q21: What is the time complexity of inserting an element at an arbitrary position in an unsorted array?

Answer: C

Explanation: Arbitrary insertion requires shifting all subsequent elements, which is O(n) in the worst case.

Q22: In a complete binary tree with n nodes, what is the height?

Answer: B

Explanation: Complete binary tree has height approximately log■(n), enabling efficient operations on heaps and balanced trees.

Q23: What is the main drawback of bubble sort?

Answer: C

Explanation: Bubble sort is notoriously slow with $O(n^2)$ complexity, making it impractical for large datasets despite being simple.

Q24: Which algorithm is generally best for small arrays of random data?

Answer: C

Explanation: Insertion Sort is simple with minimal overhead, performing well on small datasets. Efficient for nearly-sorted data too.

Q25: Which principle emphasizes hiding internal details of an object?

Answer: C

Explanation: Encapsulation hides internal implementation details behind public interfaces using access modifiers.

Q26: What keyword is used in Java to inherit from a class?

Answer: B

Explanation: The 'extends' keyword makes a class inherit from another class. 'implements' is for interfaces.

Q27: Can an abstract class be instantiated?

Answer: B

Explanation: Abstract classes cannot be instantiated directly. They serve as blueprints for concrete subclasses.

Q28: What is method overloading?

Answer: B

Explanation: Method overloading allows multiple methods with the same name but different parameter types/counts.

Q29: What is method overriding?

Answer: A

Explanation: Overriding occurs when a subclass redefines a parent class method with the same signature, enabling polymorphism.

Q30: What is the 'super' keyword used for?

Answer: B

Explanation: 'super' refers to the parent class, allowing access to parent methods and constructors.

Q31: An interface can have concrete method implementations (Java 8+). True or False?

Answer: B

Explanation: Java 8+ allows 'default' methods with implementations and 'static' methods in interfaces.

Q32: Which SOLID principle advocates depending on abstractions, not concrete classes?

Answer: C

Explanation: Dependency Inversion Principle states depend on abstractions (interfaces/abstract classes), not concrete implementations.

Q33: What is the purpose of the 'this' keyword?

Answer: B

Explanation: 'this' refers to the current object instance, useful for distinguishing local variables from instance variables.

Q34: A class can implement multiple interfaces. True or False?

Answer: B

Explanation: A class can implement multiple interfaces, separated by commas: class MyClass implements Interface1, Interface2

Q35: What is the correct order of OOP principles from least to most restrictive access?

Answer: D

Explanation: Most to least restrictive: private (none), protected (subclasses), default/package (same package), public (all).

Q36: What is a static method?

Answer: B

Explanation: Static methods belong to the class itself, not to instances. Called as ClassName.staticMethod(), not object.method().

Q37: What does REST stand for?

Answer: B

Explanation: REST is an architectural style for building web services using HTTP methods (GET, POST, PUT, DELETE).

Q38: Which HTTP method is idempotent and should be used to retrieve data?

Answer: C

Explanation: GET retrieves data without modifying it. Multiple identical GET requests produce the same result (idempotent).

Q39: What is the purpose of CSS in web development?

Answer: C

Explanation: CSS (Cascading Style Sheets) controls the visual presentation: colors, fonts, spacing, layout.

Q40: In Android, which lifecycle method is called first?

Answer: B

Explanation: onCreate() is called when the activity is first created, before onStart() and onResume().

Q41: What is the difference between an Intent and an Intent Filter?

Answer: B

Explanation: Intents request actions. Intent Filters in AndroidManifest.xml declare which intents an activity can respond to.

Q42: How do you pass data between activities in Android?

Answer: B

Explanation: Use putExtra() to attach data to an Intent, retrieve it in the new activity with getIntent().getStringExtra() etc.

Q43: What is a Fragment in Android?

Answer: A

Explanation: Fragments are reusable UI components that can be combined in activities, useful for responsive multi-pane layouts.

Q44: What is JSON primarily used for in web services?

Answer: C

Explanation: JSON (JavaScript Object Notation) is a lightweight data format for transferring structured data between services.

Q45: What does DOM (Document Object Model) represent?

Answer: B

Explanation: DOM represents HTML/XML documents as a tree of objects that JavaScript can manipulate.

Q46: What is responsive web design?

Answer: B

Explanation: Responsive design uses flexible layouts and media queries to adapt to different devices and screen sizes.

Q47: What is the purpose of a web framework (like Spring, Django, Laravel)?

Answer: B

Explanation: Frameworks provide architectural patterns, libraries, and tools to accelerate web development.

Q48: What is cross-site scripting (XSS)?

Answer: B

Explanation: XSS allows attackers to inject malicious scripts into pages viewed by other users, compromising security.